# Topic Name: Deep Learning

Deep learning refers sometimes to training large neural networks.

# Common Applications of Deep Learning

| Input(x) | Output (y) | Application |
|---|---|---|
| Home features | Price | Real Estate |
| Ad, user info | Click on ad? (0/1) | Online Advertising |
| Image | Object (1,...,1000) | Photo tagging |
| Audio | Text transcript | Speech recognition |
| English | Chinese | Machine translation |
| Image, Radar info | Position of other cars | Autonomous driving |

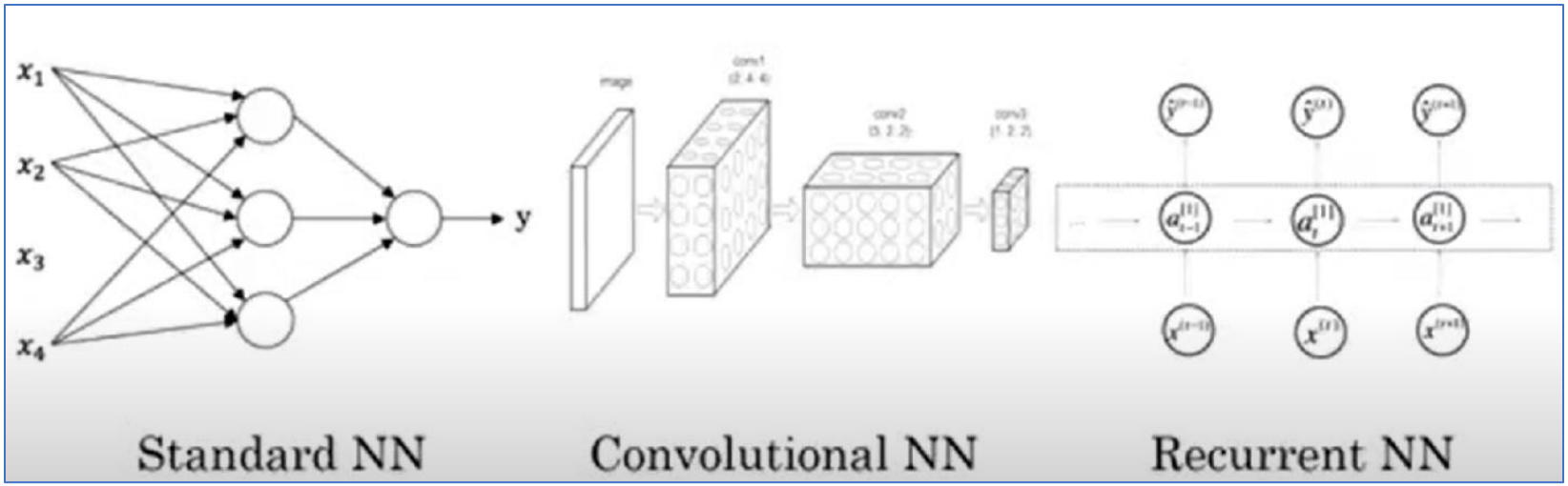
- There are slightly 20 different types of NN which are useful for different applications.

For example:

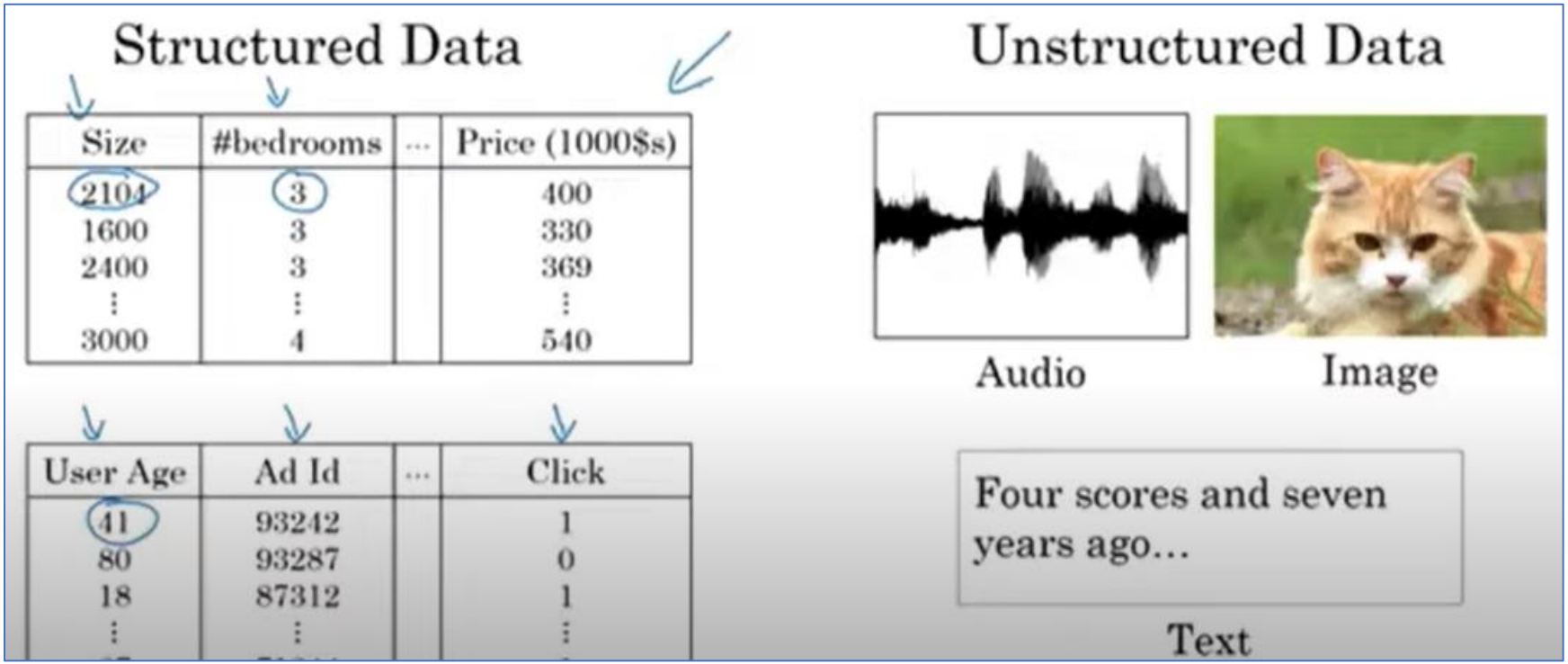
- Real estate and online advertising (standard NN)
- Images (CNN)
- Speech recognition and machine translation (RNN)
- Autonomous driving (complex hybrid NN)

CNN: Convolutional Neural Networks

RNN: Recurrent Neural Networks

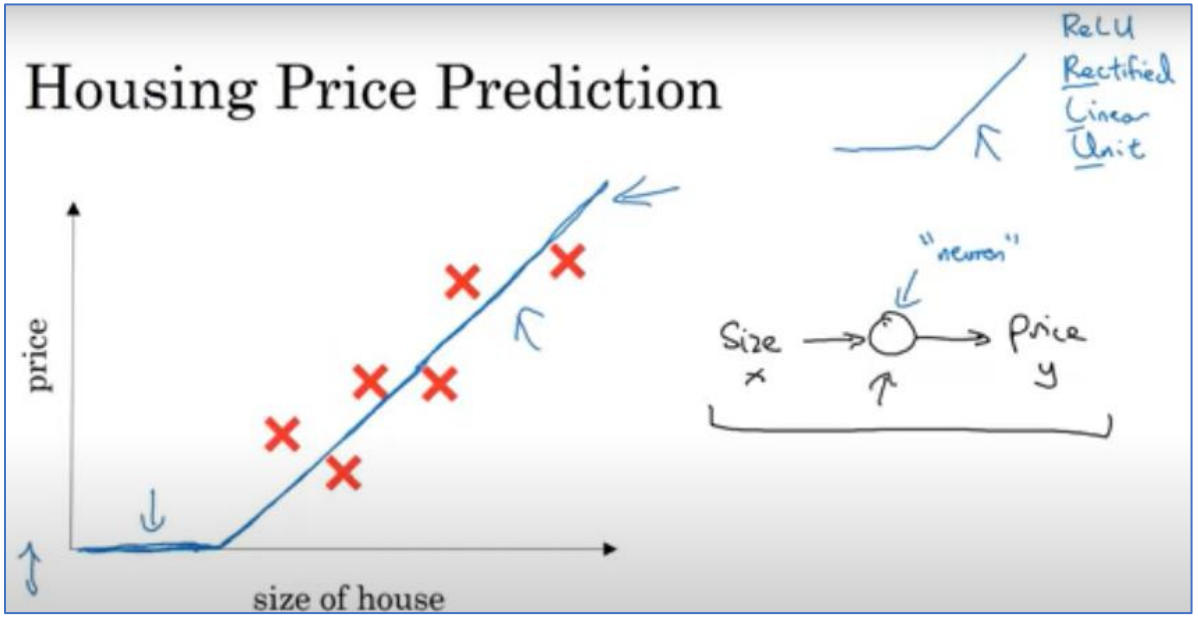

$x_1$
$x_2$
$x_3$
$x_4$
y
Standard NN
Convolutional NN
Recurrent NN

# Structured Data vs. Unstructured Data



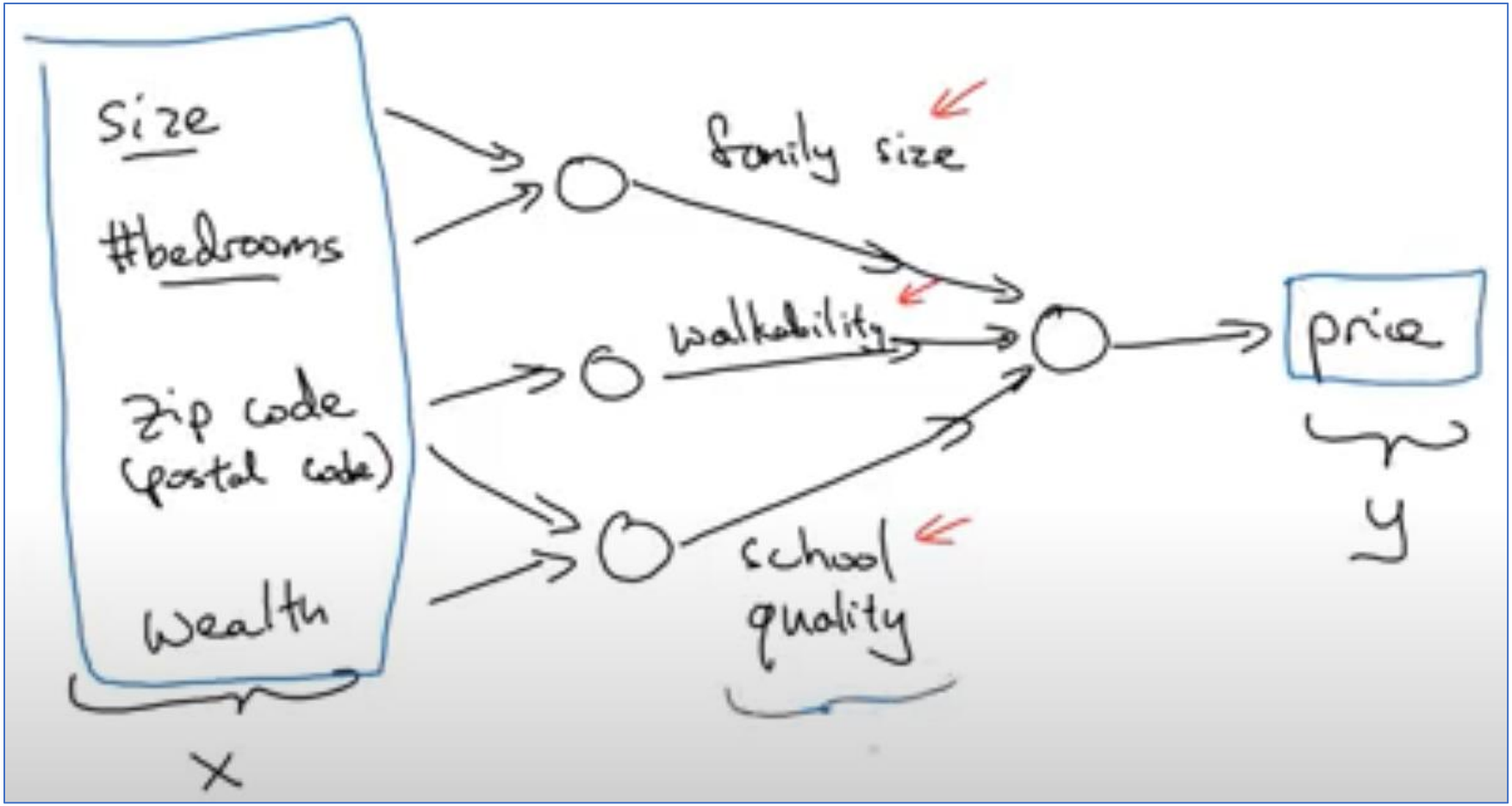

Historically it is much harder for computers to make sense of unstructured data compared to structured data.

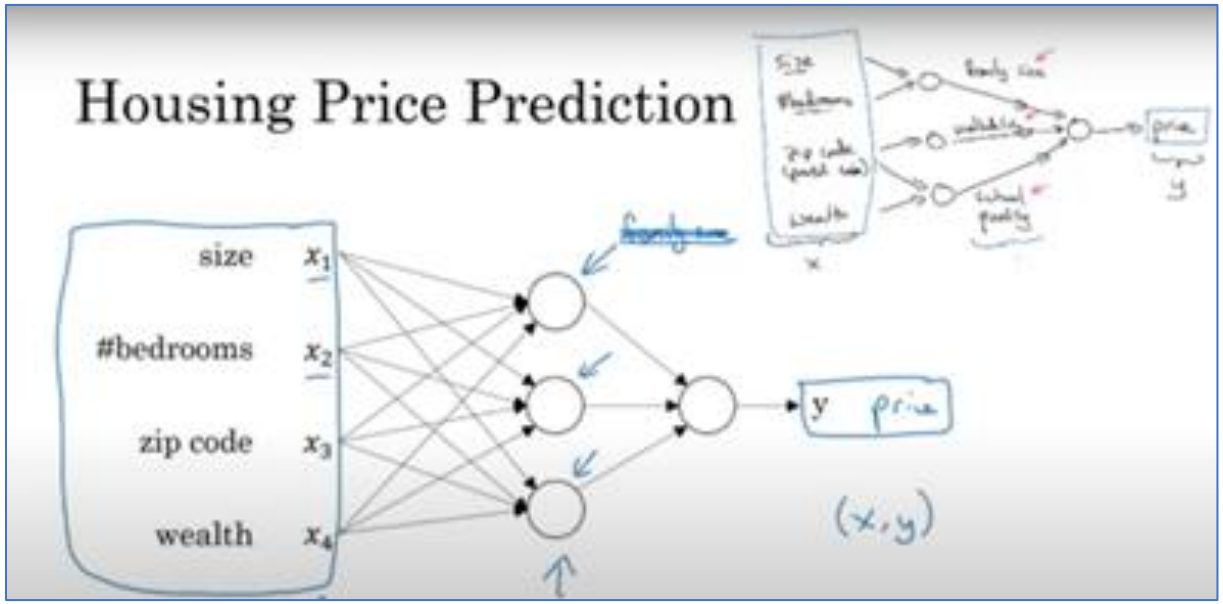
Example




RELU: Rectified Linear Unit

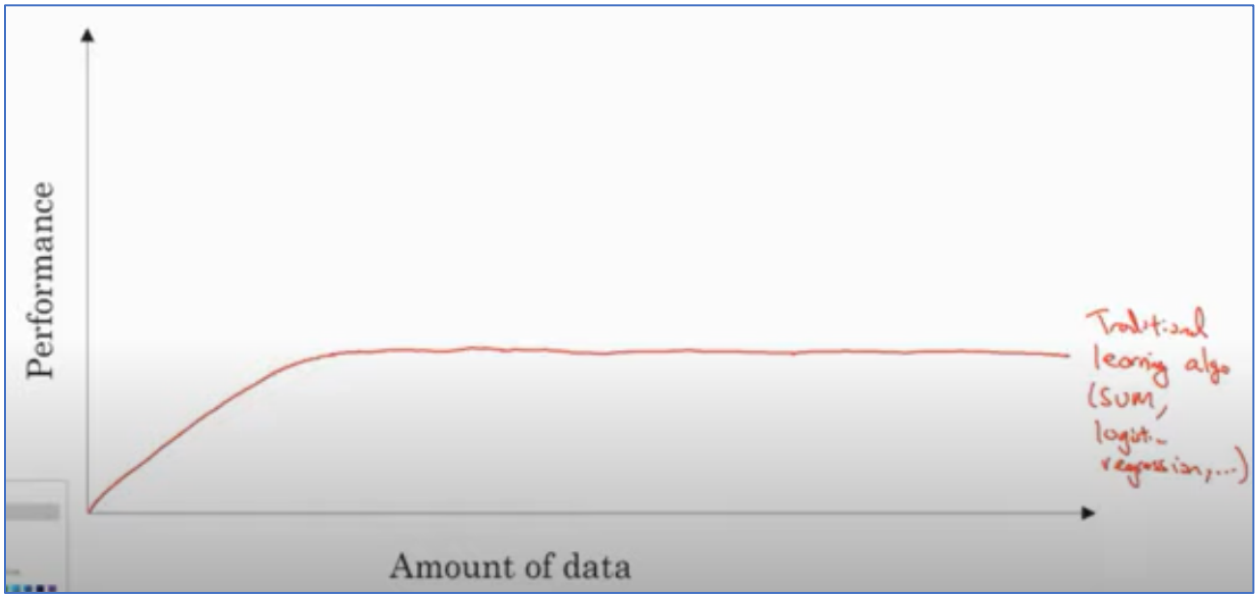
A larger neural network is formed by taking many of these single neurons and stacking them together.



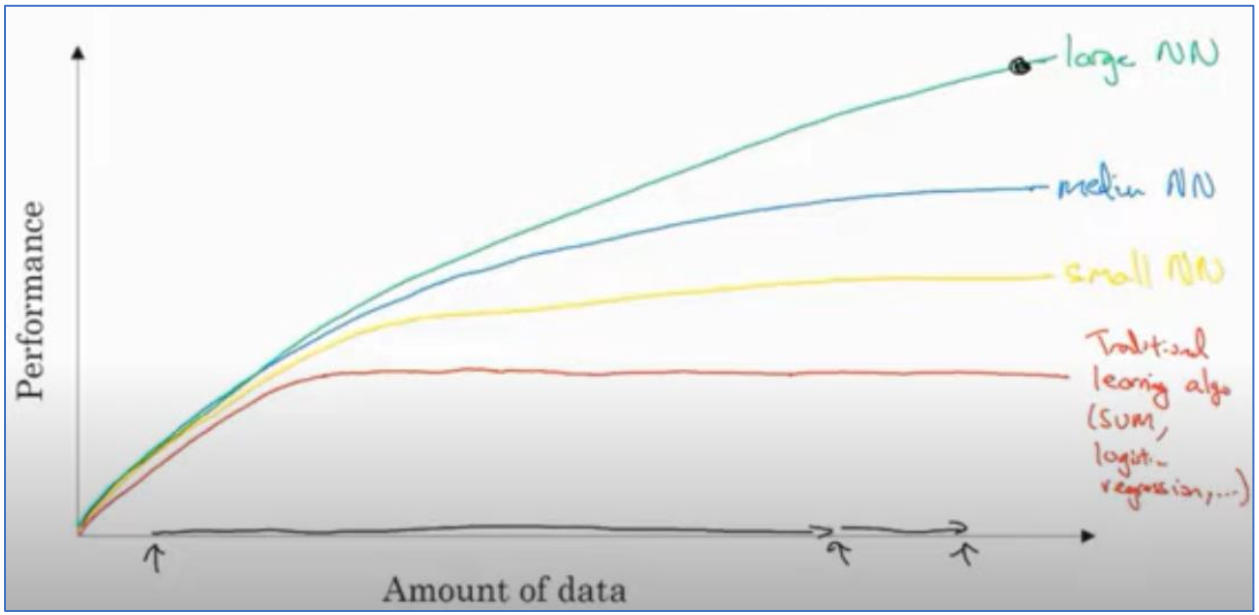



Densely connected NN

# The ideas that stand behind using deep learning these days




Traditional machine learning algorithms


Performance
large NN
medium NN
small NN
Traditional learning algo (SVM, logistic regression, ...)
Amount of data

Many reasons force us to go from a relatively small amount of data to having a large amount of data:

- Digitization of a society (we spend so much time on the computers, websites, mobile apps and activities on digital devices).
- The rise of inexpensive cameras built into our cell phones.

- All sorts of sensors in the Internet of Things.